

**DECENTRALIZED DIGITAL CURRENCY PROTOCOL**

**DDCP Reserve Management — Design  
Specification**

*Working Paper · v20260525-2*

DDCP Foundation, Inc.

May 2026

## Executive Summary

The Decentralized Digital Currency Protocol (DDCP) enables independent operators to deploy fully-reserved, fiat-pegged stablecoins in any jurisdiction where doing so is not prohibited. This document specifies the reserve design for DDCP Stablecoins: the asset composition, mandate hierarchy, supply integrity controls, and risk management framework that govern the reserves backing every DDCP stablecoin deployment.

Reserve design for DDCP rests on three structural pillars:

- **Par maintenance:** Every DDCP Stablecoin deployment must support full redemption at par at all times and under plausible stress scenarios. This is the non-negotiable primary mandate of the Reserve Management Foundation (RMF). No yield target, distribution commitment, or operational expense overrides it.
- **Political independence:** The reserve basket is designed so that no single government, central bank, or regulated entity can unilaterally impair the reserve or coerce the reserve manager into compliance actions that would compromise the value preservation mandate, to the extent permitted by jurisdictional regulators. This mirrors the adversarial distribution logic that governs the DDCP validator network.
- **Supply integrity:** Token supply is cryptographically linked to attested reserve assets. Minting requires co-signatures from both the Issuer and the RMF, and a valid reserve attestation showing that circulating supply will not exceed attested reserves after the mint. Redemption releases reserve assets only after the corresponding on-chain burn is confirmed. These controls are architectural properties, not policy commitments.

This document covers DDCP Stablecoins: fiat-pegged token deployments whose operative mandate is par maintenance. Instruments designed for purchasing power preservation rather than fiat par maintenance are outside the scope of this specification.

## 1. DDCP Design Constraints Relevant to Reserve Design

Several core DDCP design decisions constrain the reserve model and must be treated as fixed parameters:

- AML and CFT compliance is handled entirely at the Financial Intermediary (FI) layer, not at the protocol level. For purposes of this document, a **Financial Intermediary** is any entity that (a) converts between fiat currency and DDCP tokens in either direction, or (b) holds, generates, or is a required co-signer of the private key material authorizing transactions from a user's wallet. The test is functional: FI status is determined by whether either condition is met, not by the entity's label.
- No protocol-level seizure operation exists. This is a design choice, not a gap.
- Validators are internationally distributed across politically adversarial jurisdictions; no single government controls a decisive majority.
- DDCP does not host on-chain credit infrastructure. DDUSD and DDEUR holders access existing DeFi ecosystems via opt-in bridges; local currency token holders access credit through local FinTech lenders at the FI layer.
- Protocol-level changes are limited in scope and require a high-threshold supermajority of validators, to be predetermined.
- Adoption requires only the absence of prohibition, not active government support.

These constraints eliminate any reserve model that requires central bank access, government cooperation, or creates organic political alignment between the reserve and a sovereign. One structured exception is acknowledged: some deployment jurisdictions may require local currency instruments as a condition of regulatory permissibility. The local currency instruments component addresses this scenario as a deployment-specific accommodation, not a default reserve composition.

## 2. Reserve Management Foundation Mandate

All reserve design, asset composition, and yield allocation decisions are governed by an explicit mandate hierarchy. The RMF operates within this hierarchy as a binding constraint, not a set of aspirational targets. Asset composition must fulfil the mandate in priority order — no secondary objective is pursued at the cost of a higher-priority one.

## 2.1 Primary Objective: Value Preservation at Par

The RMF's first and overriding obligation is to maintain the value of every DDCP Stablecoin deployment at par with its base currency unit. This means the reserve basket must, at all times and under plausible stress scenarios, support full redemption of outstanding tokens at par. No yield target, distribution commitment, or fee obligation overrides this requirement.

Value preservation requires:

- Sufficient liquidity in the basket to meet daily redemption demand without forced sale of illiquid assets.
- Basket composition that is robust to the failure of any single component — no single asset class or jurisdiction can impair the peg.
- Conservative collateral standards that prioritize capital protection over return maximization.
- An insurance reserve, funded from yield, that provides a buffer against correlated asset shocks, custody failures, or exceptional redemption events. The reserve insurance is sized across two independent dimensions: token holder stress scenarios — correlated basket shocks, custody failures, and redemption spikes — and Operator-Issuer protection scenarios in which RMF investment underperformance within authorized risk parameters causes reserve value to fall below par. Both dimensions are modeled independently; the higher output governs the floor.
- **Open mint-and-burn as the market peg anchor:** the RMF ensures that both the token issuance window and the redemption window remain continuously open through the FI layer. A token trading above par creates an immediate arbitrage opportunity — any participant can request a mint at par through an FI and sell at market price — which constrains sustained premium above par. A token trading below par creates the inverse arbitrage — buy below par, redeem at par — which constrains sustained discount. The RMF's liquidity obligation is

to ensure the redemption window is always open; the open mint window is sufficient to prevent a sustained market price above par. Token supply expands through minting against new reserve deposits and contracts through redemption and burn — the two processes are symmetric and self-balancing by design.

- **Liquidity Float:** As outstanding issuance volume (OIV) grows, the absolute daily variability in secondary market supply and demand grows proportionally. Exchanges, market-makers, and liquidity providers require an increasingly large token float to facilitate trading at scale without routing every transaction through the mint/burn process. The Operator-Issuer is responsible for maintaining adequate Liquidity Float scaled to current OIV; if not maintained proactively, the deficit is recovered from the Operator-Issuer's Token Deployment Projects allocation before any other distribution in that tier. See §2.2 for the waterfall treatment of this obligation.

**Mandate note:** Value preservation at par is not a passive constraint — it is an active design objective. Basket composition, rebalancing triggers, and liquidity thresholds must all be specified with this objective as the binding first condition. Insurance reserve adequacy is a governance-monitored metric with defined minimum thresholds.

## 2.2 Secondary Objective: Yield Generation and Distribution

Subject to the primary objective being met, the basket is designed to generate yield from income-producing components — principally the short-duration sovereign debt sleeve and, to a lesser degree, commodity index exposure. Yield generation is a secondary objective: the basket is not optimized for maximum yield; it is optimized for value preservation with yield as a residual benefit.

Variable reserve management costs — custody fees and proof-of-reserve attestation — are recovered from gross yield at actual cost before any distribution. These are cost pass-throughs, not waterfall allocations.

Yield remaining after cost recovery is distributed according to the following waterfall, in priority order:

| Priority | Recipient / Purpose                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | <b>RMF and DDCP Foundation Management Fees</b> | Fixed operational costs of the RMF and the DDCP Foundation: legal entity maintenance, multi-signature governance infrastructure, Foundation administrative operations, and minimum governance process execution. Each deployment carries a pre-established proportional allocation. If yield after variable cost recovery is insufficient, the Operator covers their deployment's proportional shortfall directly — the Operator backstop ensures continuity of RMF and Foundation operations.                             |
| 2        | <b>Insurance Reserve</b>                       | A defined minimum share of remaining yield is directed to the insurance reserve before any distribution to Token Deployment Projects. The reserve insurance is sized against stress scenarios including correlated basket shocks, custody failures, redemption spikes, and a temporary freeze at the largest single custodian jurisdiction. Contributions continue until the reserve reaches its governance-defined adequacy threshold; thereafter, excess yield flows to Priority 3.                                      |
| 3        | <b>Token Deployment Projects</b>               | All remaining yield after Priorities 1 and 2 is distributed to Token Deployment Projects — encompassing protocol ecosystem participants and the Operator-Issuer yield allocation — at governance-defined percentages established in the RMF governance framework. The Operator-Issuer is required to maintain adequate Liquidity Float scaled to current OIV (see §2.1); if not maintained proactively, the deficit is recovered from the Operator-Issuer's allocation within this tier before any remaining distribution. |

**Yield note:** The yield waterfall is sequential — each tier is fully funded before the next receives any allocation. In periods of low basket yield (e.g., near-zero interest rate environments), lower-priority tiers may receive little or nothing. This is expected behavior, not a structural failure. The insurance reserve and value preservation mandate remain binding regardless of the yield environment.

The protocol has no built-in yield distribution mechanism to token holders. Yield flows through the RMF waterfall to Operators. Where permitted by applicable regulation, Operators may independently fund token holder incentive programs from their own yield allocation; such arrangements are at the Operator's cost and do not involve the reserve or the RMF waterfall. For DDUUSD, yield passthrough to token holders is not permitted under current GENIUS Act regulatory interpretation.

**Yield Passthrough Open Question — Relevant If CLARITY Act Is Not Enacted**

If the CLARITY Act is not enacted, the following interpretive question remains open under the GENIUS Act. Three commenters on the OCC rulemaking record support the interpretation that §4(a)(11) of the GENIUS Act targets direct PPSI payments to token holders only, not third-party arrangements: Anchorage Digital Bank (legislative drafting claim), J.W. Verret / George Mason Law (statutory analysis), and Andreessen Horowitz (policy argument). Under this reading, yield flowing to DDUSD holders through DeFi protocols (Aave, Morpho, Kamino) would not constitute a prohibited payment by DDUSD, LLC. This question should be raised in the pre-application engagement with OCC licensing staff, with record citations. If confirmed, the yield waterfall design and GTM are materially affected.

## 2.3 Supply Integrity: Mint and Burn Authorization

Token supply is cryptographically linked to attested reserve assets through four co-required controls that make supply integrity an architectural property rather than a policy commitment:

- **Dual co-signature:** The mint function requires co-signatures from both the Issuer and the RMF. Neither party can mint unilaterally.
- **Attestation prerequisite:** A valid current RMF reserve attestation is a protocol-level prerequisite for the mint function to execute. A mint that would cause circulating supply to exceed attested reserve assets cannot execute.
- **Burn-before-release:** The RMF will not release reserve assets in response to a redemption until the corresponding on-chain burn is confirmed. Supply cannot exceed reserve, and reserve cannot be drawn down without corresponding supply reduction.
- **Key management security:** The co-signature keys held by the Issuer and the RMF are managed under institution-grade security standards, including multi-signature (multisig) or multi-party computation (MPC) key management protocols, as specified in the deployment security standard. No single person or device holds sufficient key material to authorize a mint or release reserve assets unilaterally.

**Supply integrity guarantee:** These controls are symmetric: circulating supply cannot exceed attested reserve assets, and reserve assets cannot be drawn down without



corresponding supply reduction. The first three are enforced at the protocol layer — not as policy commitments subject to discretion, but as architectural constraints that cannot be bypassed by any single party. Key management security ensures that no single party can unilaterally assemble the co-signature required to circumvent them.

**Programmatic Minting Controls — Regulatory Best Practice Alignment**

The Network Firm LLP (TNF) specifically recommended to the OCC that issuers implement programmatic on-chain controls that verify reserve adequacy before permitting new token issuance, citing the October 2025 Paxos incident as the illustrative failure that on-chain controls would have prevented. DDCP’s architecture already implements this design through three co-required controls: Issuer and RMF co-signature for mint execution; a valid current RMF reserve attestation as a protocol-level prerequisite for the mint function; and confirmed on-chain burn before reserve release on redemption. If the OCC endorses programmatic minting controls as best practice in the final rule, DDCP’s architecture will be visibly ahead of that standard at the time of the PPSI application.

3. Recommended Reserve Model

3.1 Basket Structure

The recommended reserve model is a diversified basket managed centrally by the DDCP Reserve Management Foundation (RMF) serving all DDCP Stablecoin deployments. The basket comprises three core components, one conditionally included component, and two local currency components serving distinct purposes. Asset class weights are governance-defined ranges calibrated to fulfil the value preservation mandate as the primary objective, with yield generation as a secondary consideration. Prescriptive target percentages are intentionally absent from this specification; they are a governance output, not a design input.

| Asset Class                   | Composition                                                                                                                                                      | Primary Role                                                                                 | Notes                                                                                                                                                                                                                                                                                     |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Short-Duration Sovereign Debt | Investment-grade sovereign debt — T-bills and short-duration bonds — from multiple adversarially selected jurisdictions. No single jurisdiction holds a dominant | Primary yield source, liquidity buffer, daily redemption support. Funds the yield waterfall. | Income contribution varies with interest rate environment. Short duration preserves redemption liquidity. Adversarial jurisdiction distribution required. Credit eligibility criteria govern instrument selection; below-threshold sovereign debt is not eligible except within the Local |



| Asset Class                     | Composition                                                                                                                                                                             | Primary Role                                                                                                                                                                           | Notes                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | share. Eligible instruments defined by governance-ratified credit eligibility criteria.                                                                                                 |                                                                                                                                                                                        | Currency Instruments component.                                                                                                                                                                                                                                                                                                                                                             |
| <b>Physical Commodities</b>     | Physical gold held in distributed vaults across adversarially selected jurisdictions. Proof-of-reserve on-chain attestation.                                                            | Primary value preservation anchor. Crisis hedge. No single regulator can impair a decisive share of physical commodity holdings, to the extent permitted by jurisdictional regulators. | No yield contribution. Role is capital preservation, not income generation. Weight reflects value preservation mandate primacy.                                                                                                                                                                                                                                                             |
| <b>Commodity Index Exposure</b> | Diversified commodity exposure (energy, base metals, agricultural) via exchange-traded commodity index instruments. Instrument share sized below single-regulator impairment threshold. | Commodity diversification. Partial yield contribution. Inflation hedge supplementing physical commodities.                                                                             | Must remain a minority position. Instrument domicile and issuer concentration are weighting constraints. Eligible instruments include traditional exchange-traded commodity ETFs and equivalent products; tokenized commodity indices and commodity ETFs with documented underlying custodian and real-world asset attestation are also eligible subject to governance ratification.        |
| <b>Digital Assets</b>           | Overcollateralized Bitcoin or equivalent. Inclusion conditional on volatility evolution and governance ratification.                                                                    | Diversification. Reduces dependence on traditional financial system infrastructure.                                                                                                    | Governance ratification required. Liquidation cascade risk managed through minority weighting. Not a primary component. Delta-neutral digital asset strategies (synthetic yield via perpetual position hedging) are excluded; basis risk and funding rate dependency are adversely correlated with crisis conditions, making them inconsistent with the primary value preservation mandate. |

| Asset Class                       | Composition                                                                                                                                                                                                                      | Primary Role                                                                                                                                                 | Notes                                                                                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Local Currency</b>             | Local currency holdings maintained to meet anticipated daily net redemption flows. Sized based on OIV and historical net flow patterns, with governance-defined sizing parameters.                                               | Operational redemption liquidity. Ensures routine net redemption demand is met without requiring liquidation of other reserve components or FX transactions. | Standard operational requirement for every deployment. Not subject to the adversarial jurisdiction distribution design — held locally by definition. Distinct from Local Currency Instruments, which serve a separate governed purpose.                            |
| <b>Local Currency Instruments</b> | Local currency debt instruments or government bonds of the deployment jurisdiction. Deployment-specific; not a default reserve composition. Applicable only where required for regulatory permissibility or monetary neutrality. | Regulatory permissibility accommodation. Monetary neutrality instrument for relevant deployments.                                                            | RMF governance ratification required per deployment. Must not constitute a majority reserve position. Political capture risk explicitly accepted as the cost of deployment permissibility. Distinct from the Local Currency operational liquidity component above. |

## 3.2 Why This Model

The basket structure described below is the design objective for DDCP Stablecoin deployments generally. In specific deployment implementations, deviations from the adversarial distribution design may be required by jurisdictional regulatory requirements mandating particular reserve instruments or custodians. Such deviations are documented as deployment-specific governance decisions and acknowledged as constraints on achieving the full independence design, managed through the RMF governance framework.

- No single jurisdiction can freeze a decisive share of reserves — the adversarial distribution logic that governs validators applies equally to reserve custody, to the extent permitted by jurisdictional regulators.

- No organic political alliance is created with any sovereign at the core basket level. Unlike bond-backed models, no state derives fiscal benefit from token demand in the sovereign debt, physical commodities, commodity index, or digital assets components.
- The mandate hierarchy is structurally encoded: physical commodities provide the non-income-bearing preservation anchor; sovereign debt generates the yield waterfall; commodity index exposure and digital assets provide diversification with managed risk profiles.
- No fractional amplification. Reserve assets are held in true custody, not on deposit — custodians earn a fee rather than a spread and cannot deploy reserve assets as loanable funds. The Local Currency component is an exception: holdings for daily redemption liquidity may be held in deposit form at regulated local financial institutions.
- Failure modes are survivable, not existential. No single custody failure, asset class impairment, or sovereign stress event cascades to full reserve impairment. Physical commodities provide a floor that survives aggressive coordinated regulatory action.
- No affiliated custodian concentration risk. No custodian in the recommended basket is affiliated with any DDCP entity, and no single custodian holds a dominant reserve share. This eliminates the circular credit risk that arises when a stablecoin issuer concentrates reserves at an affiliated institution. The Network Firm LLP (TNF) independently identified affiliated custodian concentration as a structural gap the GENIUS Act does not adequately address on the same OCC rulemaking record as DDCP's comment. DDCP's distributed multi-custodian architecture is the architectural solution to exactly the risk TNF identifies.

### 3.3 DDUSD-Specific Reserve Constraints

For DDUSD operating under a GENIUS Act Permitted Payment Stablecoin Issuer (PPSI) license, permissible reserve assets are primarily US-dollar instruments as defined by OCC-approved categories. The adversarial jurisdiction distribution of the recommended basket applies to DDCP stablecoin deployments generally; it does not apply to the DDUSD reserve in the same form, where the GENIUS Act constrains reserve composition to OCC-approved instruments.

DDUSD's primary differentiators versus centralized stablecoins are architectural — no protocol-level freeze function and no administrative

access that could enable one, and tiered transaction privacy. Reserve design is largely determined by GENIUS Act compliance requirements for DDUSD specifically, though the adversarial distribution of reserve custody across multiple US and non-US custodians adds a resilience dimension not present in competitors whose reserves are concentrated in a small number of US custodians.

Reserve composition reporting for DDUSD complies with applicable OCC disclosure requirements. For any tokenized reserve instruments held by DDUSD, disclosure covers both the blockchain of record and the underlying custodian of the real-world asset, consistent with applicable regulatory guidance. DDUSD, LLC has the capability to comply with lawful orders through co-signature authority and FI coordination.

## 4. Reserve Management Foundation Governance

The RMF is the most sensitive centralization point in the DDCP architecture. Its governance design applies the adversarial distribution principle to reserve custody: custodians are selected across jurisdictions such that no single government can freeze a decisive share of any deployment's reserves. RMF governance presence — the jurisdictions in which RMF signatories are based — expands incrementally from Switzerland at formation to Singapore and UAE as deployment scale permits. These are distinct dimensions: custody distribution is achievable across many jurisdictions from the outset of each deployment; governance presence expands with operational maturity. Together they target the same outcome: no single government can unilaterally compromise the reserve.

### 4.1 Structural Requirements

- **Legal structure:** Foundation under neutral jurisdiction (Switzerland for initial formation; Singapore as Stage 2+ candidate), not a corporate entity subject to shareholder control.
- **Authorization:** Multi-signature structure requiring approval from representatives across multiple politically adversarial jurisdictions for any material reserve action, to the extent permitted by jurisdictional regulators.
- **Transparency:** Full on-chain proof-of-reserve attestation at defined intervals, targeting AT-C 205 examination standard or equivalent as the attestation baseline. All rebalancing transactions publicly

observable on-chain. The annual budget for AT-C 205 examination-level attestation varies depending on reserve complexity, custodian count, and reporting cadence; first-year engagements carry a significant onboarding premium. These costs are recovered as variable reserve management costs before any waterfall allocations, on a per-deployment basis.

- **Reserve composition reporting:** Each instrument identified by applicable security-level identifier (FIGI or jurisdiction-equivalent), custodian LEI or jurisdiction-equivalent entity identifier, maturity, market value, and country of custody, enabling independent verification of custody distribution. For any tokenized reserve instruments, disclosure covers both the blockchain of record and the underlying custodian of the real-world asset.
- **Conflict-of-interest framework:** A formal framework governing disclosure, recusal, and independent review for RMF signatories and governance participants with financial interests in any DDCP entity. The framework applies to: appointment and on an ongoing annual basis; votes where a personal financial interest in the outcome exists; and decisions that affect the yield waterfall allocation of any entity in which a signatory holds a financial stake.
- **Rebalancing rules:** Publicly specified rebalancing triggers with narrow discretionary bands. Yield waterfall distribution rules formally specified and governance-ratified on a per-deployment basis.
- **Insurance coverage framework:** The RMF's custodian relationship framework addresses insurance coverage for each custodian relationship on a jurisdiction-by-jurisdiction basis. Primary coverage lines: crime/specie coverage for reserve assets held at each custodian; cyber/Tech E&O coverage for the protocol layer and key management infrastructure; professional liability; and D&O. Appropriate coverage type and limits are highly fact-specific based on custody arrangement, key storage method, and technology stack. For DDUSD specifically, coverage details are disclosed to the OCC as a supervisory matter; public disclosure of coverage limits is not required and is inadvisable.

**Governance priority:** The RMF governance specification — including insurance reserve adequacy threshold, yield waterfall distribution rules, local instrument sizing methodology, and the conflict-of-interest framework — is the second most consequential governance

design decision in DDCP after the validator network itself. It must be fully specified before the first reserve deployment.

## 4.2 Ring-Fencing Architecture and Fiduciary Independence

Reserve assets are held in true custody, not on deposit — custodians earn a fee rather than a spread and cannot deploy reserve assets as loanable funds. The exception is Local Currency holdings maintained for daily redemption liquidity, which may be held in deposit form at regulated local financial institutions consistent with applicable jurisdictional requirements.

The RMF manages each deployment's reserve as a ring-fenced pool within orphaned bankruptcy-remote Special Purpose Vehicles. Reserve assets are separated by deployment: there is no pooling of assets across token instances. The management function is centralized — custody relationships, rebalancing execution, proof-of-reserve attestation, governance oversight, and the yield waterfall framework — but each deployment's assets are separately held, separately attested, and separately reported.

This separation reflects the structural incompatibility of different deployments' liabilities, prevents cross-contamination of reserve quality across deployments, and enables per-deployment yield waterfall flexibility.

Each deployment's reserve pool is associated exclusively with that deployment and its token holders. The insolvency, litigation, or legal proceedings of one deployment have no path to the reserve assets of any other deployment, because each deployment is ring-fenced with no cross-claim on another deployment's ring-fenced pool. This isolation is structural, not contractual: it follows from the per-deployment architecture in which each token deployment requires a separate legal entity holding its own authorization, its own reserve pool, and its own on-chain token contract.

Within each deployment, reserve assets are held by the RMF in its capacity as independent fiduciary manager, separate from the Issuer's own assets. A creditor, litigant, or insolvency administrator proceeding against the Issuer has no claim on the reserve pool, which is ring-fenced in the RMF's custody for the exclusive benefit of token holders. This separation is the mechanism by which the par maintenance guarantee is made credible at the legal layer, independent of the Issuer's financial condition.



For DDUSD specifically, the GENIUS Act provides that payment stablecoin holders hold a priority claim on the reserve assets backing their tokens, senior to all commercial creditors of the Issuer. The ring-fenced pool architecture ensures that this statutory protection operates cleanly: the assets backing DDUSD are unambiguously identified, independently attested, and not commingled with any other deployment's reserves.

The RMF operates as an independent fiduciary manager structurally separate from each deployment's Issuer. Each deployment's reserve assets are attested and reported independently. The Issuer holds no claim on reserve assets beyond the right to receive redemption proceeds on behalf of token holders.

#### **Comparison to Stablecoin Reserve Trust Models**

DDCP's RMF structure provides stronger holder protection than the Stablecoin Reserve Trust model pioneered by Anchorage Digital Bank. Under that model, the same entity serves as both grantor and trustee of the reserve trusts, meaning the issuer retains legal control over reserve assets in its fiduciary capacity. Under DDCP's model, the RMF is an independent fiduciary manager structurally separate from the Issuer. This separation eliminates the conflict-of-interest vector inherent in a same-entity grantor-trustee structure and provides stronger ring-fencing across multi-deployment reserve pools.

### **4.3 Open Questions for Resolution Before First Third-Party Operator Deployment**

The following items require resolution before the first third-party Operator deployment is authorized. They are recorded here as open questions for the RMF governance specification workstream.

#### **Liability premium calibration methodology.**

The inputs (deployment jurisdiction regulatory intensity, reserve composition risk profile, Operator-Issuer balance sheet depth), the floor basis-point quantum, and the governance process for recalibrating it over time as deployment risk profiles change.

#### **Insurance reserve sizing.**

The stress scenarios to be modeled include: token holder stress scenarios (correlated basket shocks, custody failures, and redemption spikes); Operator-Issuer protection scenarios in which RMF investment



underperformance within authorized risk parameters causes reserve value to fall below par; and a temporary freeze at the largest single custodian jurisdiction, where remaining liquid assets held outside the affected jurisdiction must cover 100% of anticipated redemption demand for a defined operational window. The confidence interval, floor calculation methodology, and relationship between the token holder floor and the Operator-Issuer buffer require specification.

### **Independent review process for deployment-level reserve committee objections.**

Provider selection criteria (independence from both RMF and Operator), timeline from objection to determination, binding scope (parameter question only), and cost allocation.

## **5. Accepted Risks and Design Boundaries**

The recommended basket design accepts the following risks explicitly, and documents the scope boundaries within which the reserve design operates. Each is recorded as a known design boundary rather than a gap, with the mitigation approach or design rationale stated.

### **5.1 Base Currency Denomination and Purchasing Power Scope**

The RMF's mandate is value preservation at par with the base currency unit, not purchasing power preservation. Coordinated global monetary expansion — in which all major reserve currency issuers expand simultaneously, as occurred during COVID-era quantitative easing — reduces the real value of the base currency and of sovereign debt instruments simultaneously, but does not impair the 1:1 nominal peg. A DDUSD holder under these conditions is in the same position as any other USD holder: the nominal value is preserved; the real value tracks USD.

Adversarial jurisdiction distribution of the reserve does not protect against this scenario and is not designed to. The reserve model maintains the peg to the base currency; it cannot alter the purchasing power of the currency being pegged to. Instruments designed for purchasing power preservation rather than fiat par maintenance are outside the scope of this specification.

The sovereign debt component of the reserve should nonetheless be monitored for duration and credit quality under extended low-real-yield or

high-inflation environments, as these affect the income contribution to the yield waterfall even if they do not threaten the nominal peg.

## **5.2 Digital Asset Liquidation Cascade**

The digital assets component of the basket is adversely correlated with crisis conditions: collateral price declines and redemption demand are positively correlated in financial crises. This risk is managed through minority weighting and governance ratification of inclusion. The risk diminishes as digital asset market depth and maturity increase. Governance retains the authority to reduce or eliminate the digital assets component if volatility metrics breach governance-defined thresholds.

## **5.3 Local Currency Instruments — Political Capture Risk**

The Local Currency Instruments component — the deployment-specific exception for regulatory permissibility or monetary neutrality — creates the governance capture risk that all other basket components are designed to avoid. Holding a sovereign's bonds as a significant reserve share creates a structural dependency on that government's credit decisions. That dependency — regardless of intent on either side — constrains the reserve manager's capacity to act independently of the sovereign's position. The weight ceiling is the structural safeguard against this dependency becoming operational leverage.

This risk is managed through a hard weight ceiling that prevents Local Currency Instruments from constituting a majority reserve position, and through RMF governance ratification requirement for inclusion in any specific deployment. It is accepted as the cost of deployment permissibility in constrained jurisdictions. The operational Local Currency component — held for daily redemption liquidity in every deployment — serves a distinct function and does not carry the same political capture risk, as it does not create credit dependency on the sovereign.

## **5.4 Custodian Jurisdiction Exposure**

Reserve custody distribution and RMF governance presence are distinct dimensions of the adversarial distribution design. Custody distribution — the selection of custodians across politically independent jurisdictions — can be implemented across many custodian jurisdictions from the outset of each deployment, independent of the pace of RMF governance expansion.

The adversarial distribution principle in this section applies to custody distribution specifically.

The reserve design does not attempt to prevent any single jurisdiction from holding a significant share of reserves. In practice, certain jurisdictions will hold meaningful reserve concentrations by virtue of market structure and asset class — sovereign debt at US or European custodians, physical gold at Swiss or Singapore vaults. The design objective is resilience to a temporary freeze at the largest single custodian jurisdiction, not prohibition of concentration.

The basket and insurance reserve are jointly calibrated so that a temporary freeze at the largest single custodian jurisdiction can be absorbed without impairing par maintenance. Remaining liquid assets held outside the affected jurisdiction, combined with the insurance reserve, must cover 100% of anticipated redemption demand for a defined operational window. A temporary, bounded freeze at one custodian jurisdiction is therefore a survivable partial impairment — not an existential event for the peg — provided the concentration calibration and insurance reserve sizing are correctly specified. This is one of the explicit stress scenarios addressed in the insurance reserve sizing open question (§4.3).

Custodian concentration limits within the RMF reserve parameter framework are calibrated to demonstrate continued full redemption capacity using only reserve assets held outside the largest single custodian relationship. The calibration standard requires that assets held outside the largest single custodian are sufficient to meet 100% of redemption demand for a defined operational window.

## 6. Summary of Findings

| Topic                        | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                  | Classification            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>RMF mandate hierarchy</b> | Value preservation at par is the primary objective. Yield is secondary, distributed via waterfall: RMF and DDCP Foundation management fees first (with Operator backstop), insurance reserve second, Token Deployment Projects third (with Liquidity Float obligation embedded). Variable custody costs are recovered prior to the waterfall as actual-cost pass-throughs. No secondary objective is pursued at the cost of the primary. | Binding design constraint |

| Topic                               | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Classification                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Supply integrity controls</b>    | Mint requires dual co-signature (Issuer + RMF) and a valid reserve attestation confirming post-mint supply will not exceed attested reserves. Redemption releases reserve assets only on confirmed on-chain burn. Co-signature keys managed under multisig or MPC key management standards. These controls are symmetric and architectural.                                                                                                                                                                                                                                          | Architectural property                                            |
| <b>Reserve model selection</b>      | Diversified basket: investment-grade short-duration sovereign debt (primary yield, governance-ratified eligibility criteria), physical commodities (primary value preservation anchor), commodity index exposure (diversification, tokenized instruments eligible subject to governance ratification), digital assets (governance-ratified, conditional, delta-neutral excluded), Local Currency (operational redemption liquidity, every deployment), Local Currency Instruments (deployment-specific governed exception). Centralized RMF; ring-fenced per-deployment asset pools. | Recommended                                                       |
| <b>Political independence</b>       | Custody distributed across adversarially selected jurisdictions. Meaningful concentration at individual custodian jurisdictions is expected and accepted given market structure. Design objective is resilience: remaining assets plus insurance reserve must cover 100% of redemption demand for a defined window under a temporary freeze at the largest single custodian jurisdiction. RMF governance presence expands incrementally (CH → SG → UAE) as scale permits.                                                                                                            | Resilience standard (survivable partial freeze)                   |
| <b>Failure mode characteristics</b> | No single failure (temporary custody freeze, asset price decline, sovereign stress) cascades to full reserve impairment. A temporary, bounded freeze at the largest single custodian jurisdiction is an explicitly modeled and sized stress scenario covered by the insurance reserve in combination with remaining liquid assets. Physical commodities provide a floor against coordinated regulatory action.                                                                                                                                                                       | Survivable partial impairments (incl. single-jurisdiction freeze) |
| <b>DDUSD reserve composition</b>    | GENIUS Act constrains DDUSD reserves to OCC-approved US-dollar instruments. Adversarial distribution of reserve custody across multiple jurisdictions adds a resilience dimension not present in competitors. DDUSD primary differentiators are architectural, not reserve-composition-based. DDUSD, LLC has the capability to comply through co-signature authority and FI coordination.                                                                                                                                                                                            | DDUSD-specific constraint                                         |
| <b>Digital assets inclusion</b>     | Conditionally included pending governance ratification. Liquidation cascade risk adversely correlated with crisis                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Conditional inclusion                                             |

| Topic                                                  | Finding                                                                                                                                                                                                                                                                                                                                                                                            | Classification                  |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|                                                        | conditions; managed through minority weighting. Delta-neutral strategies excluded on mandate-consistency grounds.                                                                                                                                                                                                                                                                                  |                                 |
| <b>Local Currency (operational)</b>                    | Standard operational requirement for every deployment. Sized to cover daily net redemption demand. May be held in deposit form at regulated local financial institutions. Distinct from Local Currency Instruments.                                                                                                                                                                                | Operational baseline            |
| <b>Local Currency Instruments (governed exception)</b> | Dual function: regulatory permissibility accommodation and monetary neutrality instrument in relevant deployments. Hard ceiling on weight per deployment. RMF governance ratification required. Political capture risk through structural credit dependency explicitly accepted.                                                                                                                   | Deployment-specific conditional |
| <b>Base currency scope</b>                             | RMF mandate is value preservation at par with the base currency unit, not purchasing power preservation. Coordinated global monetary expansion does not impair the 1:1 nominal peg; it is a shared macro condition of holding the base currency denomination. Instruments designed for purchasing power preservation rather than fiat par maintenance are outside the scope of this specification. | Design boundary                 |
| <b>Fiduciary independence</b>                          | RMF operates as an independent fiduciary manager structurally separate from each deployment's Issuer. Inter-deployment ring-fencing is structural: no cross-claim path exists between deployments. Issuer insolvency proceedings cannot reach reserve assets. Same-entity grantor-trustee risk is eliminated.                                                                                      | Design advantage documented     |
| <b>RMF governance specification</b>                    | Legal structure, multi-sig composition, AT-C 205 attestation baseline, reporting architecture, rebalancing triggers, conflict-of-interest framework, and governance scope must be fully specified before first reserve deployment. Three open questions identified in §4.3 require resolution before first third-party Operator deployment.                                                        | Open workstream                 |

*This document reflects analysis through May 2026 and is scoped to DDCP Stablecoins. Reserve model parameters, jurisdiction eligibility, and asset class weight ranges are subject to revision by the DDCP governance process. For the DDCP protocol architecture, validator governance design, and Financial Intermediary compliance framework, see the DDCP Architecture Technical Overview working paper.*